Parameters of Disks (AlMasry - Appendix B)

Introduction

- The most important disk parameter is the time required to locate an arbitrary disk block, given its block address, and then to transfer the block between the disk and a main memory buffer.

- This is the random access time for accessing a disk block

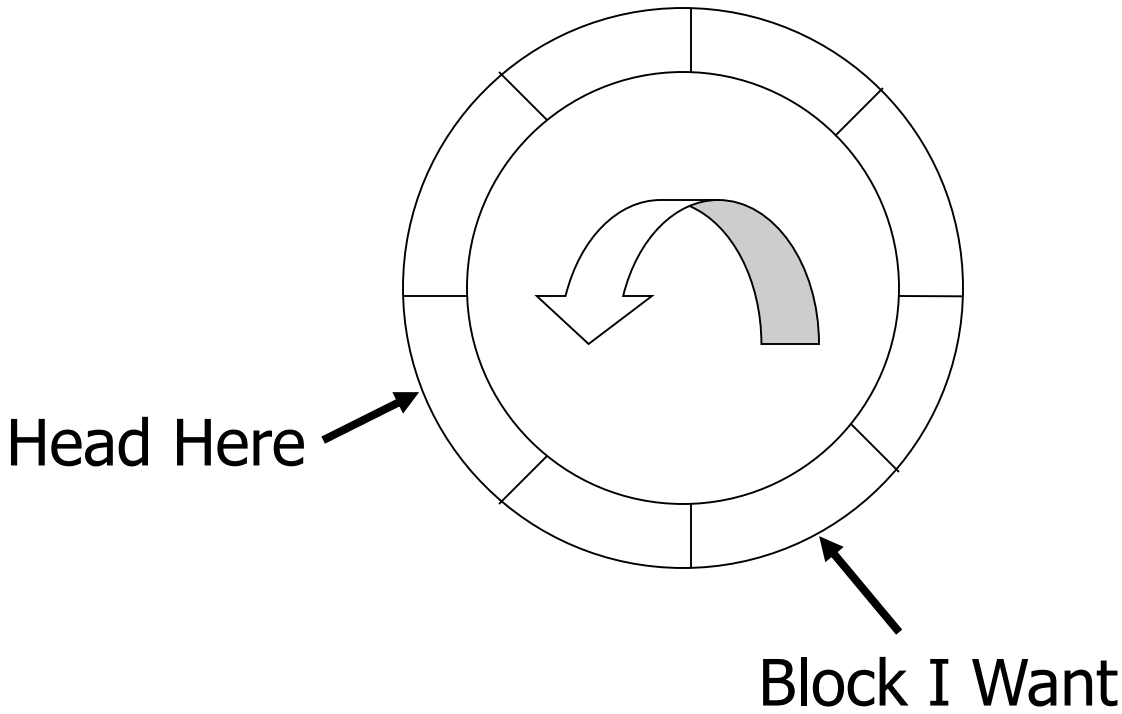
There are three time components to consider:

- Seek time (s)
- Rotational delay (rd)
- Block transfer time(btt)

Seek Time (s)

- Time needed to mechanically position the read/write head on the correct track for movable-head disks.
- Usually, the disk manufacturer provides an average seek time in milliseconds. The typical range of average seek time is 4 to 10 msec.
- This is the main culprit for the delay involved in transferring blocks between disk and memory.

Rotational Delay (*rd*)



Rotational Delay (*rd*)

- Once the read/write head is at the correct track, the user must wait for the beginning of the required block to rotate into position under the read/write head.

- On average, this takes about the time for half a revolution of the disk, but it actually ranges from immediate access to a full disk revolution.

Rotational Delay (*rd*)

- If the speed of disk rotation is p revolutions per minute (rpm), then the average rotational delay rd is given by

$$rd = (\frac{1}{2}) * \left(\frac{1}{p}\right) \min = \frac{60 \ * 1000}{2 * p} msec = \frac{30000}{p} msec$$

- A typical value for p is 10,000 rpm, which gives a rotational delay of *rd = 3 msec*.

Block transfer time (*btt*)

- Once the read/write head is at the beginning of the required block, some time is needed to transfer the data in the block.

- This block transfer time depends on the block size, track size, and rotational speed.

- If the transfer rate for the disk is tr bytes/msec and the block size is B bytes, then

$$btt = \frac{B}{tr} msec$$

Block transfer time (*btt*)

- If we have a track size of 50 Kbytes and p is 3600 rpm, then the transfer rate in bytes/msec is

$$\mathrm{tr} = (50 * 1000)/(60 * 1000/3600) = 3000 \text{ bytes/msec}$$

$$tr = \frac{50*1000}{\frac{60*1000}{3600}} = 3000 \text{ bytes/msec}$$

- In this case, btt = B/3000 msec, where B is the block size in bytes.

Total time

The average time (s) needed to find and transfer a block, given its block address, is estimated by

$$(s + rd + btt) \text{ msec}$$

- This holds for either reading or writing a block.

- The principal method of reducing this time is to <u>transfer several blocks</u> that are stored on one or more tracks of the same cylinder; then the seek time is required for the first block only.

Blocks on same cylinder

- To transfer consecutively k noncontiguous blocks that are on the same cylinder, we need approximately

$$s + (k * (rd + btt)) \text{ msec}$$

Consecutive blocks

The transfer time per block is reduced even further when consecutive blocks on the same track or cylinder are transferred.

This eliminates the rotational delay for all but the first block, so the estimate for transferring k consecutive blocks is

$$s + rd + (k * btt) \text{ msec}$$